

Zachary Rowan Copley

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Professional Experience

Product Engineer and writer, freelance

January 2024 – present

- Wrote a [report](#) on the state of AI in science search engines for Tribe AI
- Started data consultancy, [Noomatic](#), and collaborated with a product design lab on product MVP
- Built semantic search [dashboard](#) for the Data Liberation Project
- Built Python + Ansible pipeline for evaluating top 50 LLMs on HuggingFace's leaderboard
- Co-developed and pitched a Vision Pro app for workflow automation on an Opentrons liquid handler

Sabbatical

February 2023 – January 2024

- Took classes on synthetic biology at a community biology lab; created bioengineered organisms
- Attended Viable Paradise fiction writers' workshop; published in anthology [New Year, New You](#)

Lead Engineer, MiraDx

June 2021 – February 2023

- Hired and led the software team of three responsible for sample tracking systems, staff dashboards, patient onboarding flows, infectious disease reporting pipelines, HIPAA compliance, and security
- Migrated software stack to AWS after discovering HIPAA non-compliance
- Initiated an external security audit and fixed 100% of vulnerabilities above medium severity
- Generalized LIMS + ETL data model to handle new covid variants, reducing the number of days before we could report to state health authorities
- Worked with UCLA scientists to productionize diagnostics model from R to Python
- Designed and built new Next.js portal for cancer diagnostic tests, systematizing patient test tracking

Software Engineer, freelance

July 2016 – June 2021

- Wrote and published a Python ETL package for Stanford Department of Medicine, which automated protected health data preparation for consumption by machine learning models
- Built an interactive dashboard of the complete human metabolome for drug discovery research + a rendering pipeline in Python for 40k objects at 7 zoom scales using Google Maps API under the hood
- Built and optimized a visualization tool for scientists in drug discovery, streamlining JS + D3.js front-end and improving visualization rendering speed by 10x
- Built a project to monitor Chinese censorship, demoed at Oslo Freedom Forum 2022
- Updated Microsoft School Data Sync demo, documenting how to use a major API update

Research Engineer, Center for Game Science at U. Washington

December 2012 – February 2015

- Core engineering team for the award-winning synthetic biology research game Nanocrafter
- Presented our game at DARPA demo days at the Pentagon

Education

Participant, Recurse Center

November 2018 – February 2019

Focuses: deep learning, natural language processing, geospatial data, computational linguistics

Bachelor of Arts, St. Mary's College of Maryland, 2012

Computer Science and Studio Art majors, cum laude; Mathematics minor

Skills

Python, Pydantic, AWS, TypeScript, Next.js, React, D3.js, Ansible, FastAPI, Pandas
AWS Certified Solutions Architect (Associate)